



Norsk Data

Product Information

Product name

COSMOS TCP/IP Gateway
for Ethernet

Product number

211185D

Description of product

To allow use of TCP/IP protocols by applications (FTP, TELNET) in a SINTRAN environment on a Ethernet local area network.

The Telnet server in COSMOS TCP/IP Gateway for Ethernet is compatible with Telnet in Berkeley Unix 4.3. It is not compatible with Telnet in Berkeley Unix 4.2.

Responds to NetMap requests from a Network Control Server (NCS).

Reason(s) for new product or version

Version D02

- The TCP/IP did not use to checksum the incoming segments/fragments
- Known errors has been corrected

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1 Prerequisites

The following things are required in order to use this product.

1.1 Hardware prerequisites

The following hardware must be present for this product to be used:

CPU type (any of the following):

- 100 CX
- 110 CX
- 120 CX
- 500
- 5000

Floating format: **Any**

Other hardware:

Product nameProduct number

Ethernet II controller

ND 110063

1.2 Software prerequisites

The following software must be installed on the system for this product to be used:

Operating system:

NameVersion/Revision

SINTRAN III L06 PatchFile >= 2000B

Other software:

Product nameProduct number

BACKUP-SYSTEM version >=H

ND-210337

USER ENVIRONMENT version >= D03

ND-210518

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1.3 Other prerequisites

The following are necessary for this product to be used.

Minimum Permanent Mass Storage:

User	Space (pages)	Number of files
SYSTEM.....	200	4
TCP-IP.....	650	16

	Minimum	Maximum
Number of segments (ND-100)	3+4*NbOfControllers.	
Number of RT descriptions	3	

Space required on segment files: **120+256*NbOfControllers** pages

Requirements regarding internal devices:

None

Requirements regarding batch processors:

None

2 Dependencies

In addition to the above listed hardware and/or software prerequisites, the following dependencies exist between this and other products. The hardware and software listed here may, for example, be necessary to use the product's full functionality.

2.1 Hardware dependencies

There are dependency relationships between this product and the following hardware:

None

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2.2 Software dependencies

There are dependency relationships between this product and the following software:

Product nameProduct number

OWS Access Server for ND-500/5000

211325Nature of dependency

The installation of the OWS Access Server for the ND-100 depends on this product to be installed..

2.3 Other dependencies

In addition to the hardware and software dependency relationships listed above, you should be aware of the following dependencies:

None.

3 Documentation

Included (distributed free with product)TitleNumber

OpenLAN Network Supervisor Guide

830107.03 ENOptional (must be purchased)TitleNumber

COSMOS TELNET/FTP User Guide

860284.02 EN

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4 Installation procedure

4.1 Selecting Internet address

- On request from the installation program you will have to assign an Internet address for each controller. If your net is isolated from other nets outside your company, you may select the address freely, i.e. using the default values given by the installation program. If your net connects to other nets, free selection of addresses may accidentally give the same address to two installations which will create severe problems. Selecting proper Internet addresses are further described in the OpenLAN Network Supervisor guide. However, the address for application of an official Internet address range is:

DDN Network Information Center
SRI International
Room EJ217
333 Ravenswood Avenue
Menlo Park, CA 94025

4.2 Automatic installation

- Log in as user SYSTEM
- Make sure to stop the TCP-IP before starting installation.
- Insert the floppy 211185D-XX-01D in the floppy-drive
@ENTER-DIRECTORY,211185,,,
- Run the installation program:
@(211185:FLOPPY-USER)IN-TCP-IP

4.3 Non-automatic installation

When the installation program has been run:

- Update the AIP-HOSTS:SYMB file with the local internet address of the host name. See the documentation for more information.
- Update the mode file that is run after a cold start (usually called HENT-MODE:MODE) by including the following line:
@MODE (TCP-IP)TCP-IP-LOAD:MODE,,,
If you had a previous version of this product, remove the line loading the product.

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- Update the mode file that is run after a warm-start (usually called LOAD-MODE:MODE) by including the following line:
`@MODE (TCP-IP)TCP-START:MODE,,`
 after XMSG and TADADM has started.
- Update the mode file that is run before stopping the machine (usually called STOP-MACHINE:MODE), by including the line:
`@MODE (TCP-IP)TCP-STOP:MODE,,`
- OWS Access Server If you have installed the OWS Access Server for ND100 (ND211297A07), include the line:
`@(OpenLAN)OWSAS-SERV START,,`
 in the LOAD-MODE:MODE file after the @START-SERVERS command, and
`@(OpenLAN)OWSAS-SERV STOP 0,,`
 in the @STOP-MACHINE:MODE file before @MODE (TCP-IP)TCP-STOP:MODE.
- Powerfail Handling
 The powerfail handler, POPWR, is an RT program that logs in on a Terminal Access Device on the local computer. POPWR will execute any mode file that does not stop TADs or XMSG. The hardcoded input and output file names are:
`(SYSTEM)PO-PWRFAIL:MODE` and `(SYSTEM)PO-PWRFAIL:OUTP`

To enable the program to be started at powerfail one must enter SINTRAN-SERVICE-PROGRAM and type:

```

DEFINE-USER-RESTART-PROGRAM  POPWR or
NEXT-USER-RESTART-PROGRAM  POPWR      if the command
LIST-USER-RESTART-PROGRAM      gives any rt program
                                   name as output.

```

Execution of the (TCP-IP)TCP-START:MODE file and any other mode files for devices that do not handle powerfail is then included in the PO-PWRFAIL:MODE file.

After the command @MODE (TCP-IP)TCP-IP-LOAD:MODE,, in the HENT-MODE file, the SINTRAN-SERVICE command:

```

@SINTRAN-SERVICE
@DEFINE-USER-RESTART-PROGRAM POPWR  YES YES YES  or
@NEXT-USER-RESTART-PROGRAM POPWR  YES YES YES    must be
included.

```

Then the files PO-PWRFAIL:MODE and PO-PWRFAIL:OUTP must be created. User RT must have read and write access to these two files. Include in the PO-PWRFAIL:MODE file: @MODE (TCP-IP)TCP-START-D:MODE,,, and the TCP/IP will be restarted after a powerfail.

The PIOC-MONITOR included on diskette 2 is meant for use by ND service personell only.

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5 Modifications

- The FTP Server may now be started on any Ethernet-II TCP-device, the first by default (i.e. Tcp-device number 0). Edit (TCP-IP)TCP-IP-LOAD-D:MODE if you want to use another controller (see @look-at segment FTSP).

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6 Errors corrected

- In previous versions of this product, the Telnet server would sometimes continue to poll connections that was disconnecting, resulting in excessive error reports. ("Illegal socket ID")
- Previous versions of this product did not handle UDP-broadcasts correct, especially incoming UDP-broadcasts were thrown away.
- Previous versions of this product did not keep complete ARP-entries for more than 3 minutes, these will now be kept for at least half an hour.
- Previous versions of this product did not always release the internal MBUFs used by the RAW protocol, when they were no longer needed. Likewise, the xmsg-resources used by the controller when an application used SLib to send data with the RAW protocol, did not get released when finished.
- In previous versions of this product, datagrams sent to local UDP-port number X would end up at local TCP-port number X if this did exist and had been created before the equally numbered UDP-port.
- If a TCP/IP controller has been terminated due to an error, (for instance an HW error), a restart of TCP/IP may lead to a SINTRAN stop on level 14 with pagefault from level 2.
The problem is corrected by a patch in SINTRAN .